

Title: Positive pressure does not efficiently repair xylem embolism after drought

Running title: Positive xylem pressure does not promote vessel repair after drought

Authors: Giovanni Bortolami^{1,2}, Kathy Steppe³, Jan Van den Bulcke^{4,5}, Maximilian Larter^{1,6}, Salma Balazadeh⁷, Matthieu Boone⁵, Iris Frederiks¹, Toon Gheyle^{4,5}, Iván Josipovic⁵, Pierre Kibleur^{4,5}, Liselotte de Ligne^{4,5}, Anai Pereira¹, Ajaree Thonglim¹, Seppe Top³, Joris Van Acker^{4,5}, Rune Vanbeylen³, Brechtje van Helden¹, Manon van Unen¹, Frederic Lens^{1,7}.

¹Naturalis Biodiversity Center, P.O. Box 9517, 2300 RA Leiden, The Netherlands.

²Plant Ecology Research Laboratory PERL, School of Architecture, Civil and Environmental Engineering ENAC, EPFL, CH-1015 Lausanne, Switzerland.

³Laboratory of Plant Ecology, Department of Plants and Crops, Faculty of Bioscience Engineering, Ghent University, B-9000 Ghent, Belgium.

⁴UGent-Woodlab, Department of Environment, Faculty of Bioscience Engineering, Ghent University, B-9000 Ghent, Belgium.

⁵UGent Centre for X-ray Tomography (UGCT), Ghent University, B-9000 Ghent, Belgium.

⁶University of Bordeaux, INRAE, BIOGECO, F-33615 Pessac, France.

⁷Institute of Biology Leiden (IBL), Plant Sciences, Leiden University, 2333 BE Leiden, The Netherlands.

Corresponding author: bortolami.giovanni.92@gmail.com

Highlight: In tomato plants, rewatering after mild to severe drought inconsistently generates xylem pressure and rarely repairs drought-induced gas blockages in the vascular system.

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Abstract

Numerous plant species can actively pressurize water in their xylem. It has been hypothesized that regular peaks in positive pressure help plants to remove xylem embolism and to replenish water reservoirs. This may happen after rewatering when plants have suffered from soil water deficit (WD) and/or following diurnal cycles at well-watered (WW) conditions. Using X-ray microcomputed tomography (μ CT), we tested these hypotheses in tomato plants by quantifying xylem functionality and internal water pools in stems of 35 plants subjected to WD intensities before and after rewatering and 14 WW plants. Positive xylem pressure was detected in only 30% of the WD plants after rewatering, with a majority of this 30% showing no or rare events of xylem embolism repair. However, all WD tomato plants restored their internal water pools after rewatering, independently of their ability to generate detectable positive xylem pressure. In WW plants, positive xylem pressure was consistently generated and allowed for repair of the few embolized vessels. The ability of tomato plants to rehydrate non-xylary parenchyma tissues after WD and to repair embolisms under WW conditions, sheds new light on alternative water-use mechanisms by herbaceous species to recover from drought.

Keywords: drought recovery, crops, xylem embolism, symplasm, X-ray microcomputed tomography, *Solanum lycopersicum*

Introduction

When plants absorb CO_2 via their stomata to photosynthesize, water evaporates from these leaf openings into the atmosphere. This leads to capillary wicking of water molecules in leaf mesophyll cell walls exposed to intercellular gas spaces, creating a gradient of tension that pulls water from the soil towards the transpiring leaves (cohesion-tension theory, Dixon and Joly, 1895). This ingenious root-to-shoot water transport pathway is mainly driven by the

energy of the sun and atmosphere, leaving the liquid xylem sap in a metastable state (at subatmospheric pressures) that can lead to gas blockage when drought imposes strong tensions in the water column (Tyree and Sperry, 1989; Schenk *et al.*, 2015). At substantial drought intensities, gas enters in the plant vascular system, causing xylem embolism that can (partly) block the water flow, and thereby leading to organ desiccation, cell death, and eventually plant mortality (Brodribb *et al.*, 2021; Mantova *et al.*, 2022). Despite this increasing threat in a world facing climate change (Greve *et al.*, 2018; Hammond *et al.*, 2022), major knowledge gaps in how plants respond to water deficit remain (Lens *et al.*, 2022; McDowell *et al.*, 2022; Zheng *et al.*, 2023), especially related to the physiological mechanisms underlying plant capacity to recover from water deficit upon rewatering. We know that plants can recover from drought after rain or rewatering by generating new xylem conduits if lethal thresholds are not reached (Lamarque *et al.*, 2018; Gauthey *et al.*, 2022). Other studies showed that embolized conduits can be refilled with water during rehydration (Nardini *et al.*, 2011; Stewart *et al.*, 2025), which requires (near) positive xylem pressures (De Baerdemaeker *et al.*, 2019; Zwieniecki and Holbrook, 2009). Indeed, several publications claimed that plants are able to recover from xylem embolism, a phenomenon commonly called “xylem repair” or “xylem refilling” (Nardini *et al.*, 2011; Hacke and Laur, 2016), either following rewatering after soil drought (Trifilò *et al.*, 2014) or at well-watered conditions during day-night cycles (Canny, 1998). However, these studies were relying on methods which were either based on indirect measurements of xylem functionality (Salleo *et al.*, 1996; Trifilò *et al.*, 2014), on detached organs (Brodersen *et al.*, 2010; Knipfer *et al.*, 2016), or using cryo-SEM observations (Buchard *et al.*, 1999; Tyree *et al.*, 1999), that were later found to be susceptible to different (but substantial) artifacts (Charrier *et al.*, 2016; Lamarque *et al.*, 2018; Cochard *et al.*, 2000). These methodological artifacts were confirmed by direct *in vivo* observations of xylem functionality (via μ CT) that demonstrated that xylem repair is not routine in woody and herbaceous species (Cochard and Delzon, 2013), and that this outcome applies for stems, leaves and roots (Lamarque *et al.*, 2018; Johnson *et al.*, 2018; Harrison Day *et al.*, 2024). However, the debate remains unresolved, as other studies have reported xylem repair following the generation of positive xylem pressure after rewatering in

1 plants subjected to soil drought (Knipfer *et al.*, 2015; Charrier *et al.*, 2016; Gleason *et al.*,
2 2017; Drobnitch *et al.*, 2021).

3 In addition to the transpiration-driven water flow, many plants can move water in the xylem
4 via positive xylem pressure (Pickard, 1981; Schenk *et al.*, 2021). Positive xylem pressure is
5 observed in winter-deciduous tree species during springtime to rehydrate their tissues
6 before bud break (Sperry *et al.*, 1987; Niu *et al.*, 2017; Hölttä *et al.*, 2018), and in herbaceous
7 species showing guttation via hydathodes at their leaf margins at the end of the night (Miller,
8 1985; Kramer and Boyer, 1995; De Swaef *et al.*, 2013). Positive xylem pressure allows plants
9 to move water and solutes to non-transpiring tissues (e.g., roots and apical meristems,
10 Palzkill and Tibbits, 1977; Weidner *et al.*, 2005), to expand newly developing cells in
11 seedlings (Cosgrove, 1986; McIntyre, 1987; Wegner, 2014), and to facilitate xylem transport
12 in submerged aquatic plants that cannot move sap and nutrients via transpiration (Pedersen,
13 1993). Finally, positive xylem pressure has been recorded during drought stress recovery,
14 along with the observation of xylem conduit repair as observed in maize, sorghum, and
15 grapevine (Knipfer *et al.*, 2015; Charrier *et al.*, 2016; Gleason *et al.*, 2017; Drobnitch *et al.*,
16 2021), and/or rehydration of living tissues (Yang *et al.*, 2012; 2015). However, the direct role
17 of positive xylem pressure in both xylem repair and tissue rehydration after drought is
18 scarcely documented and poorly understood (Schenk *et al.*, 2021).

19 Positive xylem pressure remains an enigmatic phenomenon in plants, despite observations
20 that go back at least a few hundred years (see Schenk *et al.*, 2021 for a historical overview).
21 After all this time, it is surprising that an overview of the occurrence of positive xylem
22 pressure across a wide taxonomic range of plants is lacking. Furthermore, the underlying
23 mechanisms behind positive xylem pressure remain largely unknown: is water movement
24 driven by osmotic exudation into xylem conduits (Kramer and Boyer, 1995), hydrostatic
25 pressure in parenchyma cells associated with conduits (Pickard, 2003), or both (Schenk *et al.*,
26 2021)? And what is the role of sugar unloading from the phloem (Hölttä *et al.*, 2006)? Is
27 positive xylem pressure developing in roots, stems, or both (Schenk *et al.*, 2021; Michaud *et al.*,
28 2024)? Also, what is the role of both positive xylem pressure and dynamic symplastic

water movement (i.e., cell to cell transport from less negative to more negative water potentials) to remove xylem embolism and refill internal water storage pools during the rehydration process (Steppe *et al.*, 2006; 2015)? Technical challenges that come along with destructive methods to measure positive xylem pressure (Grossenbacher, 1938; Steudle *et al.*, 1993) have impeded us from providing answers, but more recent non-destructive techniques (Clearwater *et al.*, 2007; De Swaef *et al.*, 2013), along with visualization techniques of xylem functionality such as X-ray microcomputed tomography (Knipfer *et al.*, 2015; Charrier *et al.*, 2016), have resurrected interest in the topic.

Here, we focus on the potential role of positive xylem pressure in repairing embolized xylem conduits and rehydrating living tissues in stems of tomato plants after drought. We therefore used high-resolution X-ray CT (μ CT) scanning to investigate the stem water relationships of tomato plants, well-known for their development of positive xylem pressure (De Swaef *et al.*, 2013). We scanned the tomato plants at different time points under well-watered (WW) conditions and different intensities of soil water deficit (strong enough to generate xylem embolism, WD) before and after rewatering, while monitoring positive xylem pressure development. Our research questions are as follows: (1) Do WD plants develop positive xylem pressure after rewatering, and does this depend on the intensity of the drought applied? (2) Are the WD plants able to recover from drought-induced loss in hydraulic conductivity by repairing embolized vessels, and what is the role of positive xylem pressure on potential repair in the xylem? (3) Which tissues rehydrate in the stems of WD plants after rewatering, and is positive xylem pressure involved, or rather an alternative process? (4) Are WW plants prone to daily xylem embolism formation/repair dynamics?

Materials and Methods

Growing plants and transfer to scanning facility

At the Institute of Biology Leiden (Leiden University, The Netherlands), tomato (*Solanum lycopersicum* cv Moneymaker) seeds were sown in Murashige and Skoog medium in multiple

batches from February to August 2023. After three weeks, the plants with the most developed hypocotyls were transferred into 3.6L cylindrical pots in a mixture of potting soil (basis biomix, Lensli® substrates, Bleiswijk, The Netherlands), vermiculite, and sand in a ratio of 25:8:2, along with 3 spoons of osmocote fertilizer. All potted plants were placed in the same growth chamber at 24 °C and 70% relative humidity, with a daily 16h-light and 8h-night cycle and watered every other day until the beginning of the experiments, when the plants were 50 days old.

All experiments were run at Ghent University's Center for X-Ray Tomography (UGCT, Belgium) in three different campaigns of one week each: in April, June, and November 2023. To reduce the negative impact of plant transport from The Netherlands to Belgium (2.5 hour drive, about 200 km), and allow them to adjust to new conditions, the 50 day-old plants were transported to the UGCT in a room with additional daylight lamps (100W Erco Optec QR111) three days before starting the first scan (see Supplementary Protocol S1 and Supplementary Fig. S1 for information about the preliminary μ CT campaigns and the importance of light to generate positive xylem pressure).

Experimental design of the μ CT scans and positive xylem pressure measurements

In total, we scanned and measured the xylem pressure in 49 individual plants (14 at WW conditions, 35 at different levels of WD conditions). To obtain a large range of WD intensities, we stopped watering the tomato plants in small subsequent batches from two to three weeks before the μ CT scanning campaigns at the UGCT, for each of the three one-week campaigns. Right before the first scan (*1st scan: stem of plants at 10 cm above soil*), stem water potential (Ψ_{stem}) was measured on a single mature leaflet covered with light-reflective plastic bags for at least 2h, using a pressure chamber (Model 1000, PMS Instrument Company, Albany, OR, USA), following the recommended procedures outlined in Rodriguez-Dominguez *et al.* (2022). A few moments before scanning, the leaves from the basal 20 cm of the stem were removed and the leaf scars were immediately sealed with super glue and parafilm. There are two technical reasons for removing the leaves from the basal stem parts:

(1) the X-ray source had to be placed as close as possible to the stem to obtain the required resolution allowing us to visualize dynamics of potential embolism formation/repair in the xylem vessels, and (2) the stems needed to fit the plants in custom-made carbon-fiber tubes (CarbonWinkel.nl, Tilburg, The Netherlands) to improve the stability and straightness of the plants when they were scanned using the Environmental Micro-CT scanner (EMCT, Dierick *et al.*, 2014). It is possible that leaf removal might have slightly increased the level of native embolism at and around the scanning zone, since vessel length in tomatoes can reach 20cm (van der Schoot and van Bel, 1989). However, this slight increase can be considered as non-significant, since our results are comparable to the vulnerability curve generated by the optical technique on non-decapitated plants (see corresponding paragraph). Likewise, only a subset of the tomato individuals showed leaves around the scanning zone, while the average level of embolism was very similar across WW plants (see results section). The μ CT scanning parameters are detailed in the next section. After the first scan, we decapitated the plant stem at 20 cm from the soil surface, wrapping its cut end with parafilm to avoid dehydration, and immediately scanned the plant again at the same stem region (*2nd scan: cut stem before rewatering*). Once out of the μ CT stage, we watered the decapitated plants several times at full pot capacity until no more water was absorbed by the soil. To avoid fast dehydration of the cut surface, we put a couple of water droplets with a syringe on the stem surface and installed a silicon tube with a pressure sensor on its top (26PCCFA6D transducer, Honeywell International, Charlotte, NC, USA; or Jumo Midas C08 manometer, Fulda, Germany) according to De Swaef *et al.* (2013). We connected the pressure sensors to a datalogger (CR1000X, Campbell Scientific Ltd, Logan, UT, USA) and recorded the amount of pressure in the water column escaping the stem surface every minute (see Supplementary Fig. S2 for the sensor installation). The third scan (*3rd scan: cut stem after rewatering*) was performed at the same stem position as the previous two scans (10 cm above soil) when the pressure sensors recorded positive xylem pressure (one to three days after rewatering, depending on the timing of detectable positive pressure across the individuals) or at the end of the one-week campaign if no positive xylem pressure was recorded (three to five days after rewatering). The highest positive xylem pressure values (in MPa) during the three days after

rewatering were taken as reference for the analyses. For a subset of plants (11 out of 49), the pressure sensor was not attached to the plants when water was observed in the silicon tube (i.e., indicating the generation of positive xylem pressure). In this case, we estimated the positive xylem pressure by using the correlation between the amount of water in the silicone tube (always measured) and the recorded pressure for the other plants that were attached to the sensor ($y=0.034x+0.0003$, $p<0.0001$, $R^2=0.9$). At the end of each campaign, we collected the scanned stem segment for each individual and stored it in 70% alcohol for light microscope sectioning and observation at a later stage.

In order to assess the impact of decapitating the stem on the embolism formation/removal dynamics, we also included μ CT data of four intact plants (for more details, see second last paragraph of this section).

μ CT scans

The plants, fitted inside the custom-made carbon-fiber tubes, were installed on the stage of the EMCT scanner to scan the stem at approximately 10 cm above the soil. The tube voltage was set to 90kV and the tube emission current to 89 μ A (target power of 8W). For every scan, a total of 1600 projections were taken over one full rotation; each projection is acquired by averaging 4 images with 110ms exposure time, which resulted in the total scan run duration of 13min. Reconstructions were semi-automated with a Python script in addition to the Octopus reconstruction software package (Vlassenbroeck *et al.*, 2007). The reconstructed data consisted of total stacks of 1093 reconstructed cross-sections with an approximated voxel size of 9x9x9 μm^3 . To test whether μ CT scanning had any effect on cell death or plant vitality, we installed one non-decapitated plant in the EMCT stage with active X-ray source for 2h (corresponding to ~9 scans), prior to the three one-week experiments. Subsequently, we brought the plant back to Naturalis laboratory where we did not observe any damage on the stem scanning point or any growth reduction for the two following months. This confirms that the scans (3 times x 13min per plant) had no significant effect on tomato stem functioning. However, it is still possible that few inner cells suffered minor (non-recorded) damages.

1 **Light microscopy**

2 In μ CT scans, we were able to identify the embolized gas-filled xylem vessels (see next
3 paragraph), appearing black, while the gray water-filled vessels were hard to distinguish from
4 the water-filled surrounding tissue (Steppe *et al.*, 2004, see Fig. 2 for examples). To identify
5 all the vessels in the scanned stem area, the stems were sectioned into 50-80 μ m cross
6 sections (Reichert sliding microtome, Vienna, Austria), stained, mounted on a glass slide
7 and photographed (AXIO Imager M2 microscope, Zeiss, Oberkochen, Germany), after which
8 all vessels were counted based on light microscopy images for each scanned individual. The
9 sections were treated according to the standard protocol described by Lens *et al.* (2005). In
10 brief, the cross-sections were submerged in household bleach for 1 minute, rinsed three
11 times in distilled water, stained for 15 seconds in an alcian blue/safranin (65-35%) solution
12 (1% alcian blue in H₂O and 1% safranin in 50% ethanol), dehydrated in increasing ethanol
13 solutions (50%, 70%, 96%), submerged twice for two minutes in a 96% ethanol solution
14 mixed with HistoClear (50-50%) (Electron Microscopy Science, Hatfield, PA, USA), and finally
15 placed in 100% HistoClear. The cross-sections were mounted on glass slides with Euparal
16 mounting medium (Agar Scientific, Stansted, UK) before taking the digital images.

17 **μ CT volume and light microscopy (LM) image analysis**

18 For every scanned tomato plant, we obtained three different μ CT volumes of the stem at 10
19 cm above the soil: 1st scan on a non-decapitated plant, 2nd scan on a decapitated plant
20 before rewatering, and 3rd scan on a decapitated plant after rewatering at the time of positive
21 xylem pressure development (or at the end of the campaign in case no positive pressure was
22 generated). By carefully looking at the alignment, size, and shape of the embolized xylem
23 vessels and other defining features in the outer stem parts, we selected one 2D virtual cross-
24 section per 3D volume that best matched the image of the LM section for each of the three
25 scan time points. By doing this, we were able to compare the dynamics of xylem embolism
26 formation-repair at the same stem position for each of the vessels in the scanned
27 individuals.

In the three selected μ CT cross-sections per individual, we: i) counted the number of gas-filled xylem vessels, ii) measured the total cross-sectional area of the stem in μm^2 , and iii) measured the hydrated area of the stem (i.e., segmenting the gray-hydrated area from any black-gas-filled area; in μm^2). In the light microscopy cross-sections, we counted the total number of xylem vessels present in every sample. We calculated the level of xylem embolism as the percentage of embolized vessels (PEV) in every scan:

$$[1] PEV_{1-3} = (n_{\mu CT} / n_{LM}) * 100$$

where $n_{\mu CT}$ is the number of embolized vessels counted in the μ CT cross-section, and n_{LM} is the total number of vessels counted in the light microscopy cross-section. The xylem vessel embolism induced by decapitating the plant (2nd scan) and the number of repaired vessels (3rd scan) were calculated as:

$$[2] \text{Percentage of decapitation induced embolism} = PEV_2 - PEV_1$$

$$[3] \text{Percentage of repaired vessels} = PEV_1 - PEV_3$$

Where PEV_1 is the percentage of embolized vessels before rewatering (1st scan), PEV_2 is the percentage of embolized vessels after cutting the plant to install the pressure sensors (2nd scan), and PEV_3 is the percentage of embolized vessels after rewatering (3rd scan). The hydrated stem area (H_A) was measured in mm^2 in every scan as:

$$[5] H_A = S_A - E_A$$

where S_A is the stem cross-sectional area and E_A is the embolized area, represented by all black (i.e. gas-filled) area within each sample, excluding the embolized xylem vessels. The relative increase in hydrated stem area was calculated as:

$$[6] \text{Relative increase in hydrated stem area} = ((H_{A3} - H_{A1}) / H_{A1}) * 100$$

Where H_{A1} is the hydrated stem cross-sectional area (i.e., the gray-hydrated area minus any black-gas-filled area) in plants before rewatering (1st scan), and H_{A3} is the hydrated stem cross-sectional area after rewatering (3rd scan). Finally, the vessel diameter from all

embolized vessels was estimated from the previously selected 2D slice from the 1st scan by measuring the area of each vessel with ImageJ and computing the equivalent circle diameter. Subsequently, by carefully comparing the 1st and the 3rd scans, we differentiated between vessels that remained gas-filled and vessels that were repaired after rewatering. This last analysis was limited to WD plants that repaired at least one embolized xylem vessel after rewatering (15 out of 35 plants).

μCT versus optical vulnerability technique

Once we obtained the level of xylem embolism before rewatering (i.e., PEV_i) in the stems of the tomato plants scanned at different WD intensities, we compared the μCT results with a previously conducted optical vulnerability experiment on different batches of tomato plants of the same age and grown under the same growth chamber conditions (see details in Bortolami *et al.*, 2024). We observed that the results obtained from μCT for xylem vessels were comparable with the level of percentage of embolized pixels (PEP) obtained with the optical technique on tomato stems at similar levels of water deficit (Supplementary Fig. S3). This result confirms that the embolized vessels on the μCT scans were drought-induced.

Experimental validation in intact (non-decapitated) plants

To assess the impact of stem capitation on the dynamics of xylem embolism and potential repair cycles, we subjected four well-watered plants ($\Psi_{\text{stem}} = -0.35$ MPa) during a day/night cycle in September 2023, without recording the positive xylem pressure. These plants were grown under the same conditions as all the other plants. After the first scan (between 13:00 and 15:00h), we wrapped the entire canopy with aluminium foil and a big dark plastic bag in order to prevent transpiration (and potential embolism formation). A second scan was carried out the following morning (between 9:00 and 11:00h) to assess the level of embolism after one night, while maintaining the imposed dark conditions. After this second scan, the corresponding stem sections were sampled, sectioned and analysed via light microscopy, and the level of xylem embolism was quantified during the day and after one night as described above.

Statistical analyses

We investigated the effect of water deficit on tomato plants' ability to generate positive xylem pressure via a chi-square (χ^2) probability test on the number of plants that were or were not generating positive pressure, and a generalized linear mixed model using the stem water potential before rewatering (Ψ_{stem}) as a fixed effect. We tested the differences in the number of embolized vessels and the hydrated stem area with two independent generalized linear mixed models using positive xylem pressure (absence or presence), watering regime (WW or WD), and rewatering (before or after) as fixed effects, and a Tukey test for the 2-by-2 comparisons. Since the decapitation induced xylem embolism was limited but not entirely absent (i.e., $3 \pm 0.4\%$ PEV), we used a generalized linear mixed model on PEV levels with cutting (before or after) as a fixed effect to verify that cutting the plant did not affect xylem embolism development ($F_{1,95}=0.16$, $p=0.69$). Moreover, we demonstrated that decapitation induced embolism was not related to decreasing Ψ_{stem} , nor increasing native PEV, it did not significantly influence the xylem vessel repairing computation, and it did also not induce a loss in xylem pressure generation and xylem repair (Supplementary Fig. S4). We tested the difference in diameter between vessels that remained gas-filled and vessels that were repaired after rewatering with a generalized linear mixed model using the vessel functionality as a fixed effect and the plant as a random effect, as multiple vessels were measured on the same virtual μ CT cross-section. The relationships between the embolism before rewatering, positive xylem pressure, stem size, the percentage of repaired xylem vessels, and the relative increase in hydrated stem area were tested using linear regression models. All the analyses were done in SAS software (SAS 9.4; SAS Institute), using the GLIMMIX, FREQ, and REG procedures.

Results

Soil water deficit (WD) decreases positive xylem pressure development

We performed measurements of positive xylem pressure after (re)watering in 14 well-watered (WW) control plants and 35 water-stressed plants at different WD intensities (ranging from the onset to almost complete xylem embolization, Fig. 1). All WW plants showed positive xylem pressure, reaching the maximum pressure peaks at 0.012 ± 0.002 MPa (average $\pm$ SE, Fig. 1), while only 31% of WD plants showed detectable positive pressure after rewatering, reaching 0.007 ± 0.001 MPa (average $\pm$ SE, Fig. 1). Ψ_{stem} measured before rewatering had a significant effect on the amount of positive pressure generated across all plants ($F_{1,47}=18.2$, $p<0.001$, Fig. 1A), and on the proportion of plants in the WW and WD batches that showed positive pressure ($X^2<0.0001$, Fig. 1B). However, Ψ_{stem} before rewatering did not affect the amount of positive xylem pressure within WD ($F_{1,33}=0.02$, $p=0.9$) nor WW plants ($F_{1,12}=0.36$ $p=0.6$).

Quantification of xylem vessel embolism and stem cross-sectional area via μ CT

Tomato stems were scanned at 10 cm above the soil using μ CT to compare the level of xylem embolism and the hydrated cross-sectional area of the stems before rewatering (referring to 1st scan in non-decapitated individuals, Fig. 2A, 2C, 2E, 2G, 2I, 2K) and after rewatering (referring to 3rd scan in decapitated individuals, Fig. 2B, 2D, 2F, 2H, 2J, 2L). We observed that for all the WW plants, the majority of the few embolized vessels were repaired in almost all (13 out of 14) individuals (Fig. 2A-D). More specifically, the embolism before rewatering in WW plants declined from an average ($\pm$ SE) of $5 \pm 1\%$ of embolized vessels (PEV) to $2 \pm 1\%$ PEV after showing detectable positive xylem pressure (Fig. 3A, 3B). To cross-validate this result, we also scanned four well-watered plants that remained intact during a day/night cycle (Supplementary Fig. S5), without recording the amount of positive xylem pressure. We observed that the PEV passed from 1.5% (during the day) to 0.8% (after one night) on average. For the water stressed plants subjected to different WD intensities ($n = 35$), we subdivided them into two subgroups: WD plants developing positive xylem pressure after rewatering ($n = 11$, colored yellow in Figs. 2, 3), and plants that did not show detectable positive pressure

1 after rewatering ($n = 24$, colored red in Figs. 2, 3). Based on the 11 WD plants showing positive
 2 xylem pressure after rewatering, five of them were able to substantially repair xylem
 3 embolism ranging from an average of $53.3 \pm 9\%$ PEV before rewatering to $20 \pm 9\%$ PEV after
 4 rewatering (Fig. 2E, 2F, 3B), while the six remaining WD plants showed no or slight differences
 5 in the number of embolized vessels before and after rewatering (Fig. 2G, 2H, yellow dots near
 6 the 1-1 line in Fig. 3B). Overall, these 11 WD plants ranged from an average of $43 \pm 6\%$ PEV
 7 before rewatering to $29 \pm 6\%$ PEV after rewatering (Fig. 3A). For the 24 WD plants where we
 8 did not record positive xylem pressure (Figs. 2I-2L, 3), no substantial differences in xylem
 9 embolism were found after rewatering: only nine were able to slightly repair xylem embolism
 10 (never more than 10% PEV, red dots below the 1:1 line in Fig. 3B), while the majority of them
 11 even showed a small increase in xylem embolism (Figs. 2I-2L, red dots above around the 1-
 12 1 line in Fig. 3B). Overall, these 24 WD plants without detectable positive xylem pressure
 13 increased their embolism levels from $43 \pm 4\%$ PEV before rewatering to $45 \pm 5\%$ PEV after
 14 rewatering (Fig. 3A). We found that only the watering regime (WW and WD) had a significant
 15 effect on PEV levels ($F_{1,94}=37$, $p<0.0001$), while neither the presence or absence of positive
 16 xylem pressure nor rewatering influenced PEV. PEV results are summarized in
 17 Supplementary Fig. S6. We noticed that decapitation did not significantly alter PEV, nor
 18 plant's capacity to develop root pressure or xylem repair (Supplementary Fig. S4). However,
 19 the decapitation induced embolism are more relevant in WD plants compared to WW plants
 20 (Supplementary Fig. S4). In terms of hydrated stem cross-sectional area (Fig. 3C, 3D), all WD
 21 plants slightly to substantially increased their hydrated stem surface upon rewatering, while
 22 most WW plants remained status quo. Indeed, WW plants showed the lowest increase,
 23 ranging from $64 \pm 2 \text{ mm}^2$ to $67 \pm 3 \text{ mm}^2$ (Fig. 3C, 3D). WD plants with detectable positive
 24 pressure after rewatering increased their hydrated stem area from $42 \pm 2 \text{ mm}^2$ before
 25 rewatering to $51 \pm 3 \text{ mm}^2$ after rewatering (Fig. 3C, 3D). Similarly, WD plants without
 26 detectable positive pressure increased their hydrated area from $42 \pm 1 \text{ mm}^2$ to $53 \pm 2 \text{ mm}^2$
 27 after rewatering (Fig. 3C, 3D). The watering regime ($F_{1,94}=50.1$, $p<0.0001$) and rewatering
 28 ($F_{1,94}=20.2$, $p<0.0001$), but not the presence of positive xylem pressure, had a significant
 29 effect on the hydrated stem area.

The dynamics of xylem vessel repair upon rewatering

By comparing the scans from all the individuals, we observed that gas embolism spread in the xylem occurs in a radial direction from the inner primary xylem vessels towards the later formed secondary xylem vessels during decreasing Ψ_{stem} (Supplementary Fig. S7). In contrast, subtle xylem vessel repair (in plants with or without positive xylem pressure) showed more complex spatial dynamics (Fig. 4). In nine WD plants (out of fifteen showing some degree of xylem vessel repair), we observed vessel repairing only in the vessels closer to the cambium while the majority of vessels positioned further away from the cambium remained embolized (as in Fig. 4A, 4B). In five other WD plants, subtle xylem embolism repair was more randomly scattered as shown in Fig. 4C, 4D. One WD plant almost completely removed the gas embolism from the xylem after rewatering, making it impossible to reconstruct the dynamic sequence of events from a gas-filled to water-filled xylem (Fig. 4E, 4F). These different spatial repair dynamics were not related to the level of embolism repair or the detection/intensity of positive pressure (Fig. 5). When comparing vessel lumen diameter between vessels that remain embolized versus vessels that were repaired after rewatering, no significant difference was observed ($F_{1,2702}=0.01$, $p=0.92$, Supplementary Fig. S8).

The potential impact of positive xylem pressure, stem size, and xylem embolism before rewatering on vessel repair and water storage refilling upon rewatering

We observed that there was no linear relationship between the capacity of plants to repair their xylem vessels and to increase their stem hydrated area upon rewatering (Fig. 5). We then analyzed several intrinsic traits that might have influenced this large variability in plant ability to recover from embolism in the stem after rewatering (Fig. 6). Higher values of detectable positive xylem pressure only slightly influenced (positively) the plant capacity to remove embolism from the xylem vessels ($p=0.04$, $R^2=0.12$, standardized regression coefficient $\beta=0.34$, Fig. 6A), but did not influence the stem hydrated area changes ($p=0.6$, Fig. 6D). Smaller stems did not influence the chance of xylem vessel repair after rewatering but decreased the capacity of the stem to rehydrate its area (Fig. 6B, 6E). Finally, higher levels

of embolism before rewatering positively influenced the relative increase in hydrated stem area but not xylem vessel repair (Fig. 6C, 6F).

Discussion

In this study, we show three main results: i) periods of water deficit (WD) strongly reduce tomato plants' capacity to generate positive xylem pressure after rewatering (Fig. 1); ii) positive xylem pressure in WD plants after rewatering is not linked with the occasionally observed xylem vessel repair (Figs. 2, 3), but WD plants are much better able to replenish their internal stem water storage, mostly in the pith region (Figs. 2, 3); iii) under well-watered (WW) conditions and sufficient light intensity, positive xylem pressure in tomato plants can almost completely repair the few embolized vessels in stems (Figs. 1-3); and iv) decapitation did not significantly affect embolism formation and repair based on comparison of scans within the same decapitated WW and WD individuals (Supplementary Fig. S4) and using μ CT observations on four non-decapitated WW individuals (Supplementary Fig. S5). These results suggest that positive xylem pressure in tomato plants can prevent the xylem from reaching the point of exponential embolism spread during WW conditions, possibly driven by atmospheric drought. Additionally, the recovery of the stem hydraulic capacitance after WD suggests water flow via the non-xylary, symplastic pathway towards the pith cells (Figs. 4-6).

Positive xylem pressure does not efficiently promote xylem embolism repair after soil WD in tomato plants

Based on our observations that soil WD strongly hampered the ability of tomato plants to develop positive xylem pressure after rewatering, we can convincingly state that positive xylem pressure is not an efficient mechanism in tomato stems to repair drought-induced xylem embolism. Out of the 35 individuals that were subjected to different WD intensities and subsequent rewatering, 11 showed detectable positive xylem pressure, of which only five individuals were able to substantially repair the xylem embolism generated by drought

at 10cm above the soil (Fig. 3). Among these five WD-subjected individuals, it is unknown whether the repaired vessels were connected to the functional root-to-shoot water transport or rather blocked by embolized vessels in upper stem parts. Our results agree with the consensus that xylem repair is not routine in herbaceous or woody plants after soil water drought (Cochard and Delzon, 2013; Johnson *et al.*, 2018; Lamarque *et al.*, 2018; Stewart *et al.*, 2026), and contradict the observations in the herbaceous monocots sorghum (Drobnitch *et al.*, 2021) and maize (Gleason *et al.*, 2017), and in the woody eudicot *Vitis* spp. (Knipfer *et al.*, 2015; Charrier *et al.*, 2016) where positive pressure was enhanced by rewatering after WD (compared to WW conditions) and associated with xylem vessel repair (Knipfer *et al.*, 2015; Charrier *et al.*, 2016). It is worth noticing that we miss μ CT scans on intact WD plants that would have nicely complemented our main results based on decapitated plants. However, since stem decapitation did not influence positive pressure generation nor xylem repair dynamics (Supplementary Fig. S4), we assume that any potential decapitation artifact did not have a major impact on the main results during the one-week μ CT experiment.

It remains an open question why we observed the variability in detectable positive xylem pressure and the associated differences in xylem repair among the WD plants after rewatering, and why the stress intensity does not directly influence the recovery capacity. This question is linked to the incomplete understanding of the mechanism(s) underlying positive xylem pressure. As reviewed by Schenk *et al.* (2021), two non-mutually exclusive mechanisms can contribute to water movement under pressure: (1) the osmotic exudation due to a gradient in osmotic water potential from the soil to the vessel-associated cells in the xylem, and/or (2) the hydraulic pressure through activation of aquaporins and H^+ -ATPase pumps. It has been recently shown that tomato plants increase suberin deposition in the cell walls of the root exodermis following WD (Cantó-Pastor *et al.*, 2024). Although we have not investigated the root anatomy, a thickened exodermis might decrease the osmotic-driven soil-to-xylem water flow even after rewatering when the soil osmotic potential is near zero. Moreover, WD might have caused an irreversible disruption of the soil-root conductance by destroying fine roots and root hairs (Koch *et al.*, 2025), with a potential subsequent loss in xylem pressure generation. Furthermore, stomatal closure in tomato plants happens very

early at the onset of drought, thereby strongly reducing photosynthesis even during mild drought conditions (Bortolami *et al.*, 2024). As WD conditions prolong, this may lead to a shortage in the movement of soluble sugars towards the roots and stems, which in turn may increase the osmotic potential (i.e., becoming less negative) in the vessel-associated cells and/or inactivate the aquaporins and H⁺-ATPase pumps in the different root compartments. Our observation that WW plants growing in darker daytime conditions generated substantially lower values of positive xylem pressure (Supplementary Fig. S1, confirmed by Papanov *et al.*, 2020) supports this hypothesis, since photosynthesis and phloem flux are light-dependent (Sellier and Mammeri, 2019; Miyoshi *et al.*, 2023). The potential role of phloem unloading in developing positive xylem pressure is consistent with its light dependency (Papanov *et al.*, 2020). Interestingly, the fact that nearly all the decapitated WW plants were able to generate positive xylem pressure after the stem-leaf continuum was disrupted, highlights the involvement of redundant mechanisms in addition to phloem unloading. In conclusion, positive xylem pressure in tomato plants is heavily impacted by soil WD due to a range of potential changes in root anatomy (e.g., suberin deposit in outer root layers), availability of soluble sugars, sufficient light conditions, and/or other unknown variables, meaning that there are several potential reasons for why rewatering does not consistently result in detectable positive xylem pressure and vessel embolism repair.

Rehydration of stem water pools after WD is substantial, and unlikely related to positive xylem pressure

Our results clearly show that the extra xylary tissue can be rehydrated more easily than xylem vessels upon rewatering WD plants (Fig. 3). The expansion of these stem tissues following rewatering agrees with studies finding a clear relationship between stem (or petiole) diameter and water potential (Bourbia and Brodribb, 2023; Peters *et al.*, 2025; Harrison Day *et al.*, 2026). This emphasizes the role of stem capacitance and internal water storage pools during tomato recovery from drought. Hydraulic capacitance in plant organs relates to the amount of water they can internally store or release per unit of water potential variation, and has been highlighted as an important functional plant trait related to drought tolerance

(Steppe *et al.*, 2007; Salomón *et al.*, 2017; Epila *et al.*, 2017; De Boeck and Steppe, 2025). This holds especially true for herbaceous plants and stem succulents like cacti that have stems with abundant living parenchyma tissues (Nobel *et al.*, 1992; Corso *et al.*, 2020; Verbeke *et al.*, 2023), and also in leaves where the decline in water content outside the xylem is an important intrinsic cue at the beginning of drought stress (Scoffoni *et al.*, 2014; 2017). However, the role of hydraulic capacitance during the recovery phase from drought remains incompletely known. Outside-xylary water is mostly thought of as an internal water reservoir that can temporarily delay water shortage periods (Gleason *et al.*, 2014; Degraeve *et al.*, 2021; Pereira *et al.*, 2024). Interestingly, we observed that all WD tomato plants were able to (almost completely) recover their stem area and rehydrate their parenchyma cells (especially in the central pith), independently of their capacity to develop detectable positive xylem pressure. Such water redistribution usually occurs at night under well-watered conditions, when cells that have been depleted during the day due to transpirational water loss are rehydrated, thereby restoring their turgor pressure, which in turn enables cell division and plant growth (Steppe *et al.*, 2006; 2015). This cell-to-cell water movement is typically governed by a water potential gradient, with water flowing from regions of less negative (or zero to positive) water potential (e.g., the xylem) toward regions of more negative water potential (e.g., extracellular spaces and living tissues, Steppe *et al.*, 2006; 2015). This mechanism of symplastic water movement is in line with our observations related to stem area rehydration and recovery observed in our study: more stressed plants also restore water pools more efficiently (Fig. 5F), confirming that more negative water potentials attract more water into gas-filled parenchyma cells. Our results mostly suggest that this symplastic water pool is not connected to the root-to-leaf water flow as plants can: i) more easily rehydrate the empty (air) space in the stems and not the xylem vessels (Fig. 5), ii) in the rare cases of vessels repairing, this is not systematically related to a precise spatial pattern (Fig. 4). Moreover, since bigger stems (and therefore bigger amount of hydrated tissue) are not related to a better vessel repair (Fig. 6B), we can exclude that the water flowing via the cell-to-cell pathway might require longer time (our experiment is based on one-week window) to

1 build up enough turgor in the cells to develop positive pressure and subsequent vessel
2 recovery.

3 Summarizing, this dynamic water exchange between xylem vessels and the surrounding
4 tissues highlights the complexity of xylary and non-xylary water pathways, especially in
5 critical survival stages as the recovery period. For instance, Stewart *et al.*, (2025) showed
6 that barnyard grass (an herbaceous monocot) can repair xylem embolism before recovery of
7 the surrounding tissues, while *Pleopeltis polypodioides* (an epiphytic resurrection fern)
8 rehydrates the nonvascular tissues before showing tracheid embolism repair (Prats and
9 Brodersen, 2021), suggesting that recovery strategies can be lineage-specific. While the
10 regular embolism spread through the xylem vessel network (from the inner to the outer
11 vessels) demonstrates that the primary xylem is more vulnerable to embolism
12 (Supplementary Fig. S7), the random spatial arrangement of the few repaired vessels
13 suggests that multiple water pathways (both symplastic and apoplastic) are involved with a
14 temporal and spatial dynamics that remains obscure (Fig. 4; Steppe *et al.*, 2012; Hubeau and
15 Steppe, 2015). For instance, the vessels that are closer to the cambium could be rehydrated
16 by the phloem via transversal rays, while the inner xylem conduits could be rehydrated by
17 water coming from the inner parenchyma cells in the pith area, and both could receive water
18 from downstream vessels that are still functional. We conclude that these dynamic water
19 movements in non-xylary tissues of herbaceous species are often ignored (Skelton *et al.*,
20 2017), and advocate that they likely play a more important role in plant survival of forbs and
21 grasses than anticipated.

22 **Positive xylem pressure repairs subtle losses in daily xylem conductivity in tomato** 23 **stems under WW conditions**

24 Although cycles of substantial embolism formation and repair on a daily basis have been
25 documented in the literature (Salleo *et al.*, 2004; Nardini *et al.*, 2011; Venturas *et al.*, 2017),
26 there is increasing evidence that widespread levels of embolism in the field are not routine
27 and mainly limited to periods of severe drought (Cochard and Delzon 2013; Delzon and
28 Cochard 2014; Lens *et al.*, 2022). This does not, however, contradict observations of mild

1 levels of xylem embolism throughout the growing season under normal natural conditions,
 2 as reported for herbaceous (< 20% loss of stem conductivity, Lens *et al.*, 2016) and various
 3 temperate tree species (< 20% loss of stem conductivity, Wason *et al.*, 2018; Guan *et al.*,
 4 2022). Indeed, we found subtle levels of xylem embolism (on average 5% of embolized
 5 vessels) in the stems of tomato plants even under controlled WW conditions. We
 6 acknowledge here that cutting leaves close to the scanning zone due to practical constraints
 7 might have slightly increased the level of native embolism in our plants during the first scan.
 8 However, when comparing the results with the optical technique, we can confirm that these
 9 changes are subtle (Supplementary Fig. S3). Moreover, the overall decapitation effect was
 10 limited, especially in WW plants (Supplementary Fig. S4), and the validation of the results by
 11 scanning intact (i.e., non-decapitated) well-watered plants during a night/day cycle confirms
 12 that plant decapitation did not significantly influence our results. After positive xylem
 13 pressure had consistently been detected in the WW individuals, the average percentage of
 14 embolized vessels slightly dropped from 5% to 2%. This indicates that positive xylem
 15 pressure does not completely repair the subtle levels of xylem embolism. Does this mean it
 16 is not relevant for a tomato plant to partly recover from these minor losses in xylem hydraulic
 17 conductivity?

18 Although we did not assess whether the few repaired vessels were actually functionally
 19 contributing to the root-to-shoot water transport system or whether the repaired vessels
 20 would be more vulnerable to recurrent embolism (Hacke *et al.* 2001; Bouda *et al.*, 2019), we
 21 argue that partly recovering from daily minor losses in xylem conductivity is important, both
 22 under declining and high soil water content. First, when soil water content decreases, subtle
 23 embolism-repair dynamics during nighttime may have important consequences for the
 24 spread of gas in the 3D conduit network. Like most vascular species, tomato plants present
 25 a sigmoidal vulnerability curve (Supplementary Fig. S3), indicating that the spread of gas in
 26 the vascular tissue exponentially increases at a certain threshold, which is defined by the
 27 pressure inducing 12% loss of hydraulic conductivity (P_{12}). In this context, even repairing 3%
 28 of the vessels might be an important avoidance strategy to delay reaching the P_{12} threshold,
 29 thereby avoiding a rapid embolism spread in the vasculature. Second, when soil water

content is sufficiently high, plants might benefit from partly repairing the embolized vessels during atmospheric water deficit periods (i.e., when the atmospheric VPD increases). It has been observed that when VPD increases (i.e., with high temperatures and low relative air humidity), xylem embolism can spread in the vascular tissue even under sufficient soil moisture (Schönbeck *et al.*, 2022). Consequently, nocturnal positive pressure peaks could be important in sensitive species by repairing the few most vulnerable vessels that embolized during the warmest hours of the previous day (Jacobsen and Pratt, 2012; Venturas *et al.*, 2017), thereby removing part of the VPD-induced embolism and preventing the vascular system from reaching P_{12} . This theory is also reinforced by our observation that after removing the day/night cycles, xylem pressure generation was heavily reduced.

Conclusions

Our results in tomato plants show that most tomato plants subjected to soil WD cannot repair embolized xylem vessels after rewatering, but they can rehydrate the inner part of their stems (pith parenchyma cells) that may or may not be linked to the transpiration pathway. We also demonstrate that positive xylem pressure can consistently repair some of the few embolized vessels that arose during the day under well-watered conditions due to evaporative demands. The combination of both (subtle) xylem vessel repair and (substantial) internal water storage replenishment in tomato stems indicates that refilling of gas-filled cells happens via two non-mutually exclusive mechanisms: (i) fine-scale adjustment of xylem conductance directly driven by positive xylem pressure (especially in WW plants, but strongly hampered in WD plants after rewatering), and (ii) substantially rehydrating non-xylary parenchyma tissues via different sources of dynamic cell-to-cell water flow (becoming the more dominant process after post-drought rewatering). These findings demonstrate how water-use strategies in vulnerable crops support drought recovery, with important implications for the agro-industry in securing tomato yields both in greenhouses and in the field.

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Author contributions

GB, FL, KS, JVdB, and ML planned and designed the research.

GB, SB, MB, IF, TG, IJ, PK, ML, LdL, AP, AT, ST, RV, JVA, BvH, and MvU performed experiments.

GB analyzed the data.

GB, FL, KS, and JVdB wrote the manuscript with contributions from all authors.

Conflict of interest

The authors declare no competing interest.

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Data availability

Collected data are available in the supplementary information, the raw μ CT volumes and light microscopy pictures can be shared by the corresponding author.

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Figures legends and alt text

Figure 1. Soil water deficit (WD) decreases the ability to develop positive xylem pressure in *S. lycopersicum*. **A)** Relationship between the Ψ_{stem} reached before rewatering (MPa) and the maximum recorded positive xylem pressure (MPa) in the three days following rewatering in plants under well-watered (WW, green dots) and water deficit (WD, orange dots) conditions. The vertical blue line indicates the point of stomatal closure (gs_{90}), the yellow, brown, and red vertical lines refer to the different levels of embolism development in the stem xylem (P_{12} , P_{50} , and P_{88} , respectively) measured via the OV technique. gs_{90} , P_{12} , P_{50} , and P_{88} values were obtained from Bortolami *et al.* (2025). **B)** Frequency of plants showing detectable positive xylem pressure under WD (orange bar) and WW (green bar) conditions. The significance of the χ^2 test is indicated.

Alt text Figure 1. Scatter plot showing water deficit reduces development of positive xylem pressure after rewatering versus well watered tomato plants, plus a bar chart of how many plants developed detectable pressure.

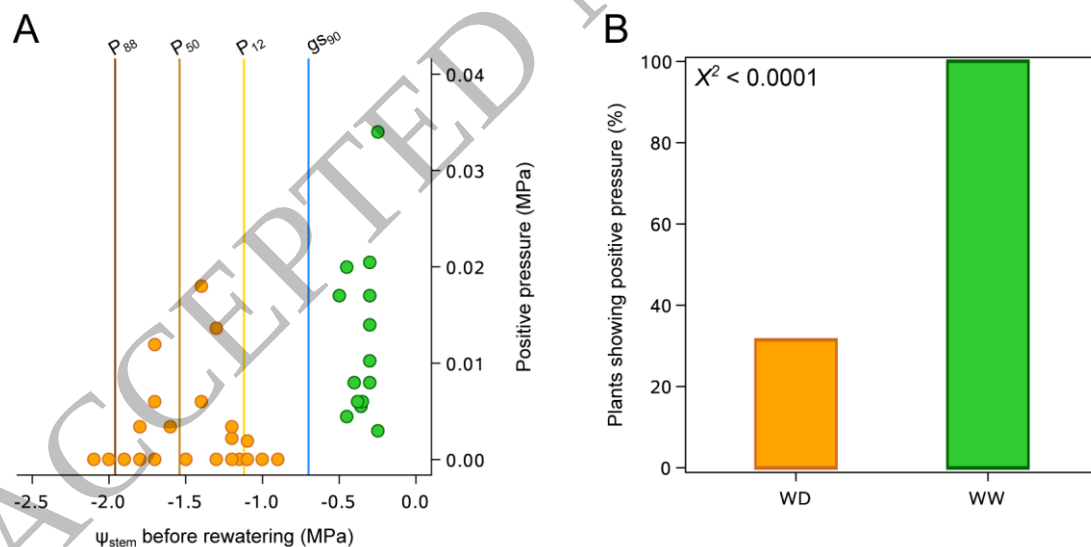


Figure 2. Cross-sections of the μ CT volume of tomato stems in six individuals before and after rewatering show different embolism repair dynamics. After the 1st μ CT scans in plants before decapitation (**A, C, E, G, I, K**), the stems were cut, rewatered, and scanned again after one to five days (**B, D, F, H, J, L**). Significant xylem embolism repair is detected in

the WW plants (green, **A-D**) and rarely in the WD plants with positive xylem pressure after rewatering (yellow, **E-H**), but never in the WD plants without detectable xylem pressure after rewatering (red, **I-L**). The selected cross-sections show the same location in the stem for the six individuals before and after rewatering (10 cm above the soil). The Ψ_{stem} (MPa) before rewatering, the amount of positive xylem pressure (P, MPa) after rewatering, the proportion of embolized vessels (PEV, %), and the hydrated stem area (mm^2) are indicated for the three individuals. Scale bars correspond to 5mm. The blue arrows indicate the xylem vessels able to change their status from gas- (black) to water-filled (grey), while pink arrows indicate xylem vessels remaining embolized after rewatering.

Alt text Figure 2. Microcomputed tomography cross sections of tomato stems before and after rewatering, showing embolism repair in well watered plants, occasional repair in water deficit plants with positive pressure, and none without positive pressure.

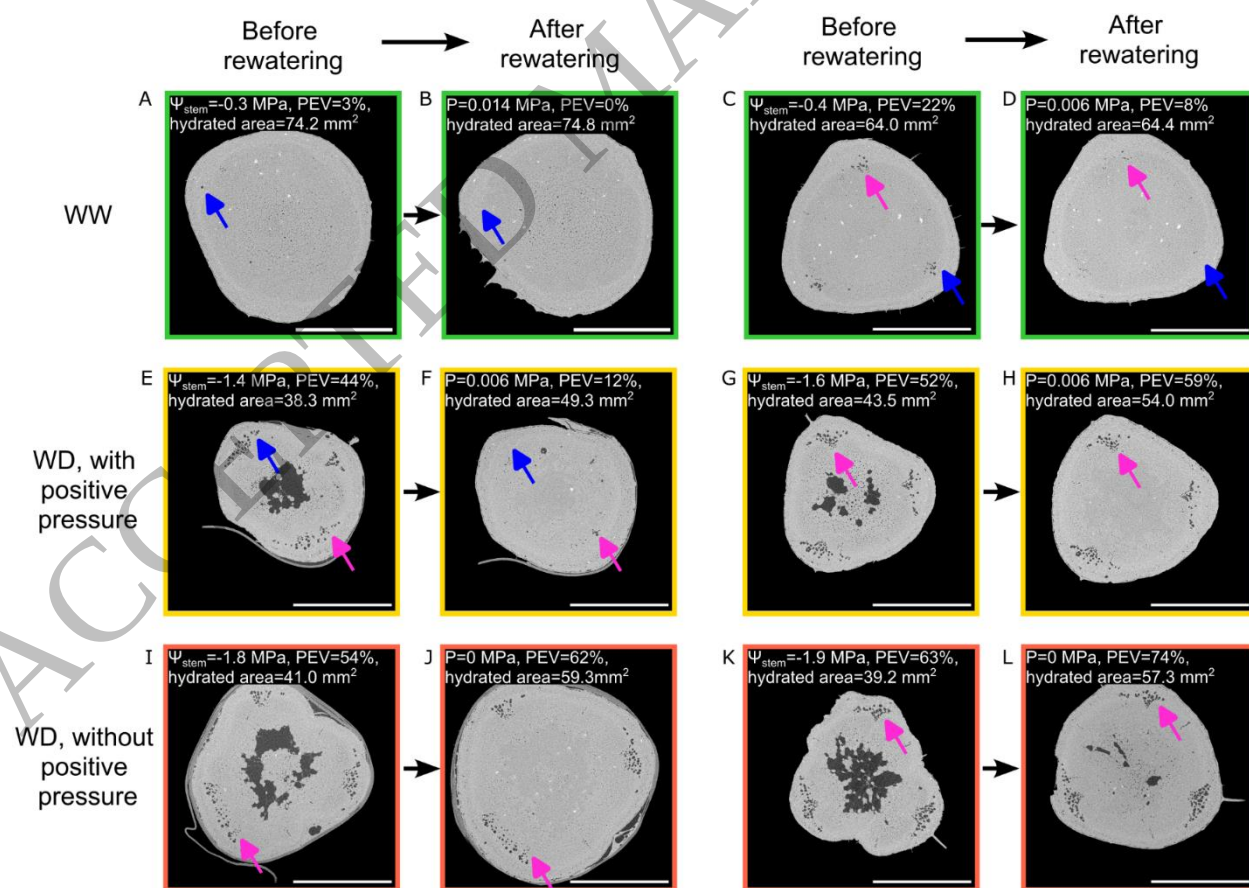


Figure 3. Embolized xylem vessels and hydrated area of tomato stems. **A)** Percentage of embolized vessels (PEV) slightly dropped in WW plants (green) and WD plants showing positive xylem pressure (yellow) after rewatering, but further increased in WD plants not developing detectable positive pressure (red). PEV was counted in the μ CT cross-sections from the 1st scans (i.e., PEV before rewatering, striped boxes) and 3rd scans (i.e., PEV after rewatering, full boxes). **B)** Relationship between PEV before and after rewatering (%) counted in the μ CT cross-sections. Symbols represent individual WW plants (green), WD plants showing positive xylem pressure (yellow), and WD plants without detectable positive pressure (red). Dots below the black 1:1 relationship line represent individuals with xylem embolism repair after rewatering, which is most prominent in some of the WD plants that were capable of developing xylem pressure. **C)** hydrated stem area (mm^2) in the μ CT cross-sections from the 1st and 3rd scans, showing a significant increase in stem area after rewatering in the WD individuals. Box colors and patterns are the same as in panel A. **D)** Relationship between the hydrated stem area before and after rewatering (mm^2) measured in the μ CT cross-sections, showing a subtle to substantial increase in stem area in the WD plants after rewatering and a status quo in most WW plants. Symbols represent individual plants with the same color code as panel B. The black line represents the 1:1 relationship. In panels A and C, boxes and bars show the medians, quartiles, and extreme values, diamonds within boxes correspond to means, and circles outside boxes correspond to outlier values.

Alt text Figure 3. Box (panels A, C) and scatter (panels B, D) plots showing that percentage of embolized vessel (panels A, B) decreased slightly after rewatering well watered plants and water deficit plants generating positive pressure, but not for water deficit plants not generating positive pressure, while hydrated stem area (panels C, D) increased for water deficit plants after rewatering, independently of their capacity to generate positive pressure.

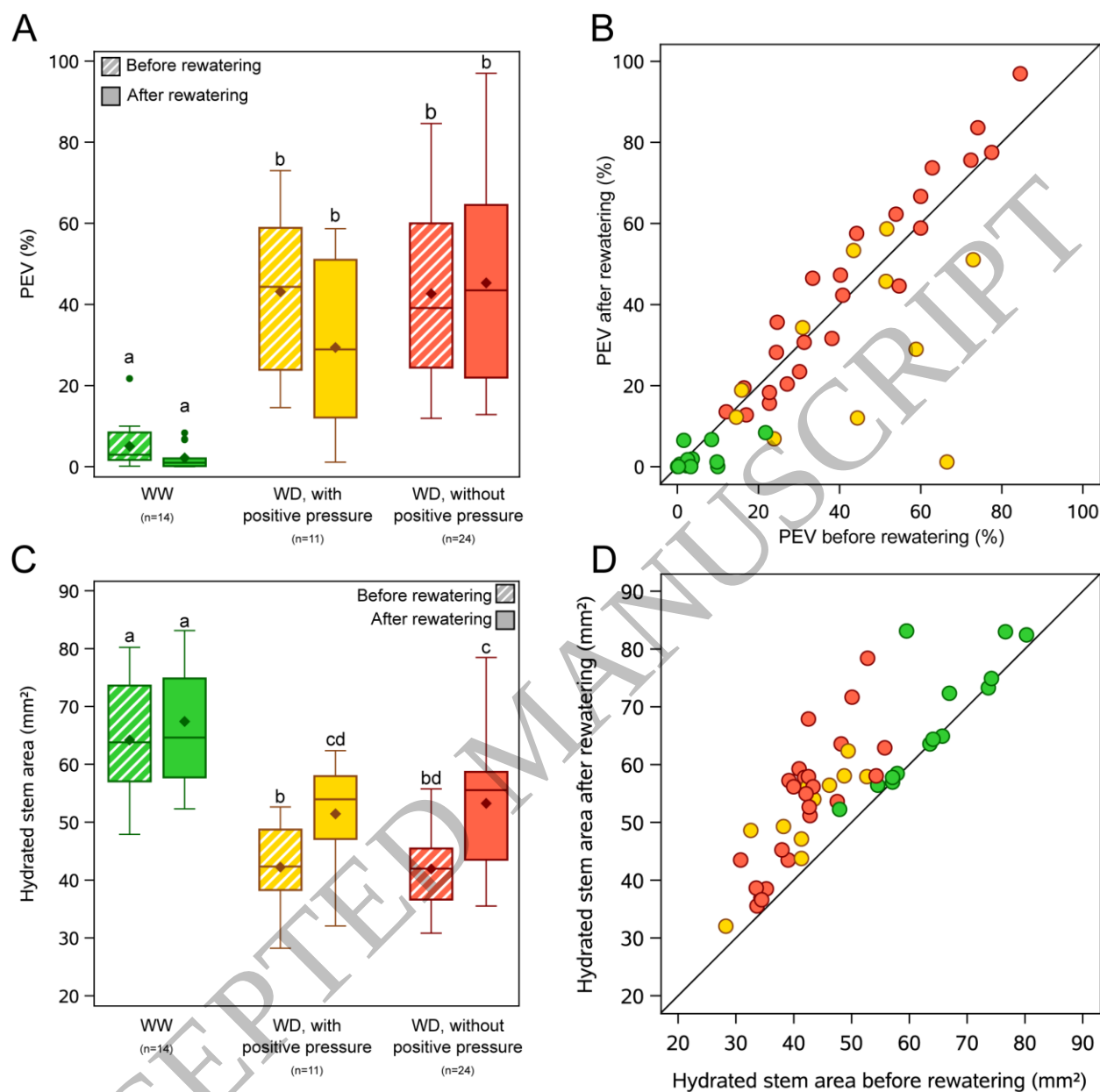


Figure 4. Dynamics of xylem embolism repair in tomato. Two-dimensional reconstructions of cross-sections from μ CT volumes of tomato stems showing repair of xylem embolism. The scans were taken at the same spot in three different plants before (A, C, E) and after (B, D, F) rewatering. The arrows indicate xylem vessels that changed their status from gas-filled (red arrows) to water-filled (blue arrows) after rewatering. Vessel repair showed roughly two different patterns among the fifteen WD plants that were able to (partly) recover from xylem embolism: most of the plants (nine) repaired only the most recently

formed vessels (**A, B**), and five plants showed a random pattern of embolism repair (**C, D**). In one remaining WD individual, xylem embolism repair was almost completed (**E, F**). The Ψ_{stem} (MPa) before rewatering, the amount of positive xylem pressure (P , MPa) after rewatering, and the amount of embolized vessels (PEV, %) are indicated for every sample. Scale bars correspond to 500 μm .

Alt text Figure 4. Microcomputed tomography cross section of tomato stems showing embolism repair patterns after rewatering, some water deficit plants repaired only recently formed vessels, others showed a more random repair pattern not related to vessel proximity to the cambium.

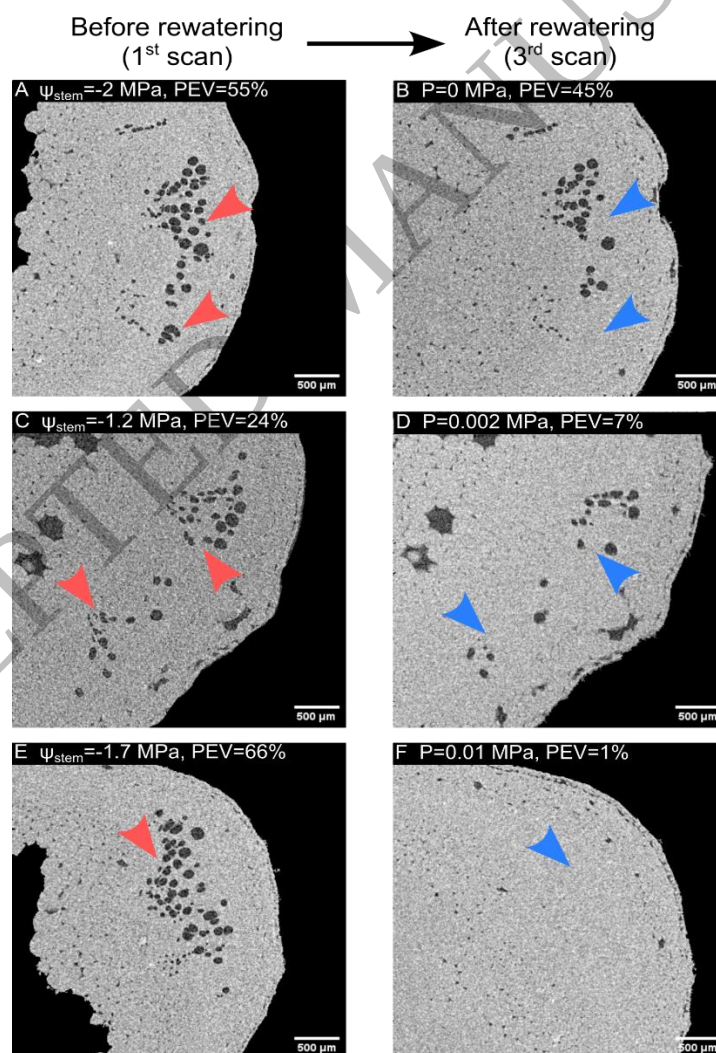


Figure 5. No link between water deficit (WD) plants' capacity to repair embolized xylem vessels and to recover their stem area. Relationship between the number of repaired vessels (% compared to the total number of vessels) and the relative increase in stem hydrated area (%) after rewatering. Dots represent single replicates, the symbol color indicates the recorded positive xylem pressure after rewatering from white (no pressure recorded) to black (maximum: 0.035 MPa).

Alt text Figure 5. Scatter plot showing no relationship between the percentage of repaired xylem vessels and the relative increase in stem hydrated area after rewatering among water deficit tomato plants.

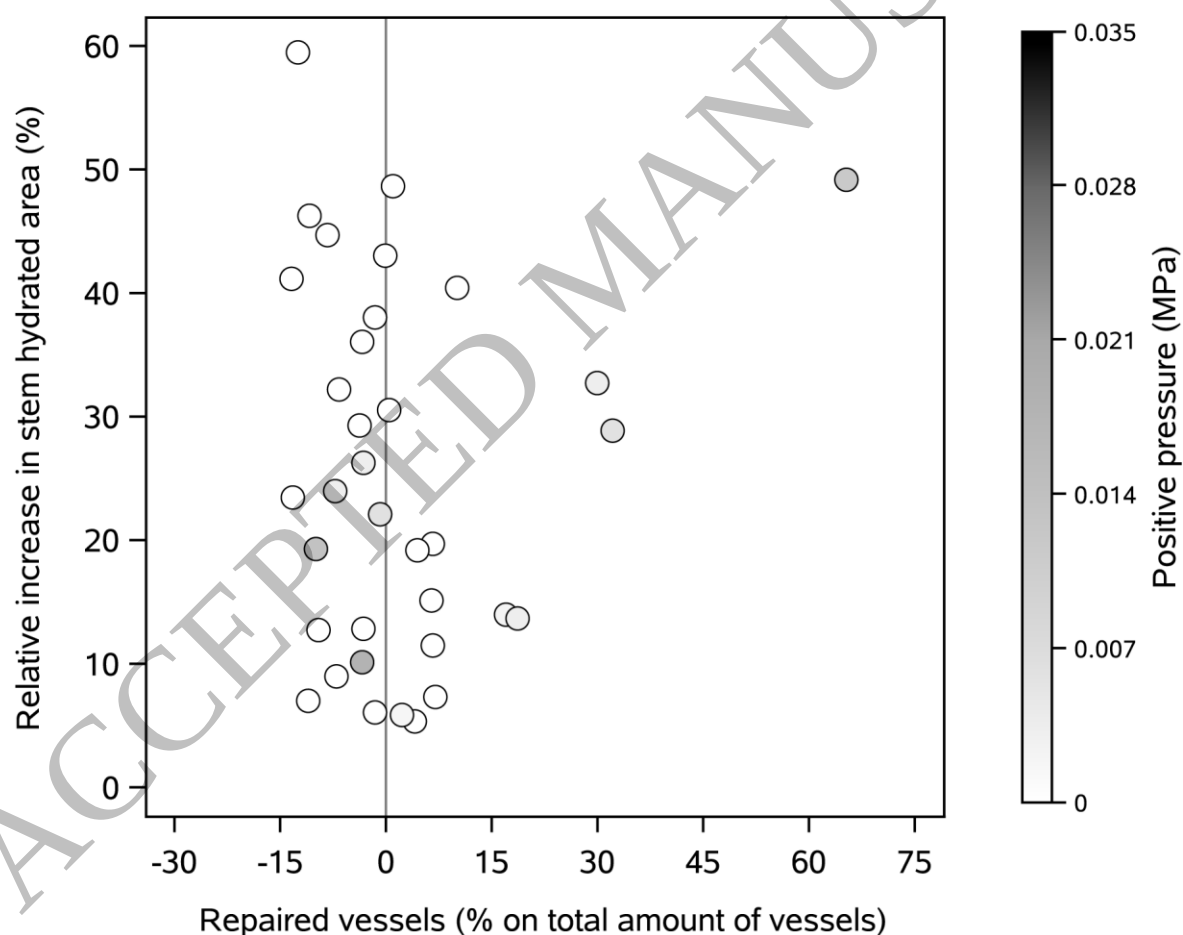
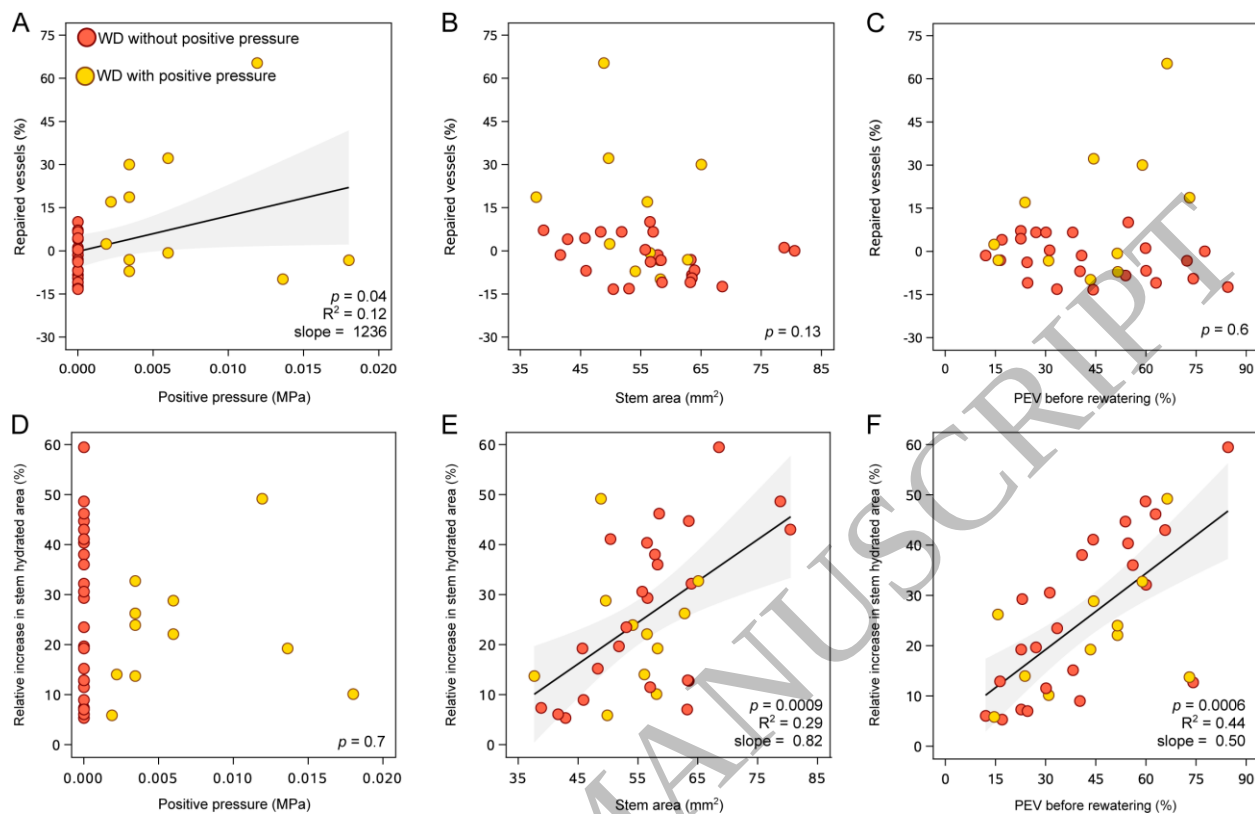


Figure 6. The potential role of three selected functional traits on vessel repair and relative increase in stem area after rewatering in water deficient (WD) plants. **A)** Positive xylem pressure (MPa) has a slight positive effect on xylem embolism repair after rewatering (number of repaired vessels compared to the total number of vessels). **B)** No correlation between stem area (mm²) and percentage of repaired vessels after rewatering. **C)** No correlation between xylem embolism before rewatering (PEV) and percentage of repaired vessels after rewatering. **D)** No correlation between positive xylem pressure (MPa) and relative increase in stem hydrated area (%). **E)** Larger stem area (mm²) is correlated with a higher relative increase in stem hydrated area after rewatering. **F)** Higher xylem embolism before rewatering (% PEV) is linked with a higher relative increase in stem hydrated area after rewatering. For every panel, symbols represent single replicates, colors represent WD plants with positive xylem pressure (yellow), and WD plants without detectable positive xylem pressure (red), lines refer to the linear regressions with shaded areas showing the 95% confidence intervals. For the significant relationships, the *p*-values, *R*², and the slope of the linear regression analyses are indicated within each panel. For the non-significant relationships, only the *p*-values are indicated.

Alt text Figure 6. Six scatter plots examining relationships among positive xylem pressure, stem area, embolism before rewatering, vessel repair, and stem area increase in water deficit plants.



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